INCH-POUND

MIL-DTL-24640/9D
22 August 2002
SUPERSEDING
MIL-C-24640/9C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 22, TYPE 2XS (INCLUDING VARIATION TYPE 2XSO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, AWG size 22.

Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance with codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.

Third - Two conductors, one black and one white, twisted together. The pitch shall be 1½ ± ¼ inch.

Fourth - Binder tape applied helically with overlap, at manufacturer's option.

Fifth - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.

Sixth - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.

Seventh - The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

Eighth - Binder tape applied helically with overlap.

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CONSTRUCTION VARIANT FOR TYPE 2XS

Ninth - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE 2XSO

Ninth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Tenth - Binder tape applied helically with overlap, at manufacturer's option.
Eleventh - Crosslinked polyolefin jacket. (See Table II for thickness).

TABLE I. Construction details.

PIN M24640/9	Type number	Number of pairs	Jacket thickness (inch) (min)	Cable Diameter (inch) (min)	Cable Diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
-01UN	2XS-2	2	0.038	0.308	0.332	61.4
-02UN	2XS-3	3	.038	.325	.350	74.2
-03UN	2XS-7	7	.041	.423	.455	135.0
-04UN	2XS-10	10	.044	.537	.579	194.0
-05UN	2XS-14	14	.044	.582	.627	245.0
-06UN	2XS-19	19	.044	.644	.694	319.0
-07UN	2XS-24	24	.047	.758	.818	431.0
-08UN	2XS-30	30	.047	.804	.866	500.0

TABLE II. Construction details - overall shield.

PIN M24640/9	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	Cable diameter (inch) (max)	Cable weight (pounds per 1000 feet) (max)
-01UO	2XSO-3	3	0.271	0.038	0.380	0.410	117.0
-02UO	2XSO-7	7	.373	.041	.474	.510	190.0
-03UO	2XSO-10	10	.487	.044	.594	.640	287.0
-04UO	2XSO-14	14	.541	.044	.636	.686	323.0

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-05UO	2XSO-19	19	.607	.044	.709	.765	423.0
-06UO	2XSO-30	30	.777	.047	.869	.937	644.0

REQUIREMENTS:

Qualification required.
Construction details: See Tables I and II.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 kHz spark test 5.5 kV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical: Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum 17.37
Voltage withstand - volts, root mean square, minimum
Conductor to conductor and shield 1000

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Conductor to shield ..... Not applicable
Shield to shield (component shield only) 200
Component shield to overall shield (type 2XSO only) 500
Insulation resistance - megohms-1000 feet, minimum
Conductor to conductor and shield 2500
Shield to shield (component shield only) 100
Shield to shield (component shield to overall shield 2XSO only) 100
Conductor and shield continuity ..... No failure
Jacket flaws (2XSO only) ..... No failure

Group A:
Visual and dimensional ..... No failure
Capacitance:
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum 50
Capacitance unbalance - percent, maximum 15
Characteristic impedance - at 1 MHz, ohms ..... 40 min, 63 max
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage) ..... Not applicable
Weight ..... No failure

Group B:
Cold bending No failure
Cross link proof test (percent maximum)
Jacket 50
Drip ..... Not applicable
Tear - pounds per inch thickness, minimum 35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum 3
Gas flame (1 hour) Not applicable

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - cycles, minimum. ..... Not applicable
Permanence of printing (jacket) - cycles, minimum 125

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Shield - conformance to material, construction and coverage .....No failure
Heat distortion - percent of unaged, maximum.....30
Cable sealant removability.....Not applicable
Shrinkage .....No failure
Mutual inductance - dB/1000 feet, maximum .....Not applicable
Surface transfer impedance (type 2XSO only)
    Milliohms per meter, maximum 700
    EMP response, dB, minimum 60
Pulse response time - ms/500 feet
    Rise time, maximum.....Not applicable
    Decay time, maximum .....Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable).....No failure
Acid gas equivalent - percent, maximum
    Jacket.....2
    Fillers.....2
    Insulation 18
Halogen content - percent, maximum
    Jacket.....0.2
    Fillers.....0.2
Flame propagation .....No failure
Hot oil immersion
    Tensile strength - percent of unaged, minimum.....50
    Elongation - percent of unaged, minimum 50
Smoke index, maximum
    Jacket.....25
    Fillers.....45
    Insulation 10
Toxicity index, maximum
    Jacket.....5
    Fillers.....5
    Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket).....No failure
Fungus resistance .....No failure
Abrasion resistance .....No failure
Gas flame (3 hour) .....Not applicable
Component wire .....No failure

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Unit ordering length: 2 - 14 500 feet nominal.
19 - 30 250 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-009)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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